



NORTHERN BEACHES SECONDARY COLLEGE

# MANLY SELECTIVE CAMPUS

## HIGHER SCHOOL CERTIFICATE

### Trial Examination

2017

## Mathematics

### General Instructions

- Reading time – 5 minutes
- Working time – 3hours
- Use **black** pen
- Write your Student Number at the top of each page
- Section I – Multiple Choice – use the Answer Sheet provided
- Section II – Free Response – use a separate booklet for **each** question.
- Board approved calculators and templates may be used.
- Reference sheet provided.

### Section I Multiple Choice

- 10 marks
- Attempt all questions

### Section II – Free Response

- Questions 11-16 – 90 marks
- Each question is of equal value
- All necessary working should be shown in every question.

Weighting: 40%

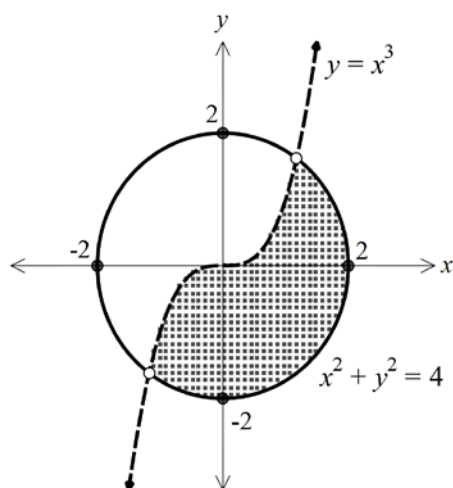
**Section 1 Multiple Choice: Attempt Questions 1 – 10**

Answer questions on the provided answer sheet.  
Allow approximately 15 minutes for this section.

Q. 1 36.1984 written in scientific notation, correct to 4 significant figures is:

- (A)  $3.620 \times 10$
- (B)  $3.62 \times 10$
- (C)  $3.620 \times 10^{-1}$
- (D)  $3.6198 \times 10^{-1}$

Q. 2 The inequalities which define the shaded region shown in the diagram are:



- (A)  $x^2 + y^2 \geq 4$  and  $y < x^3$
- (B)  $x^2 + y^2 \geq 4$  and  $y \geq x^3$
- (C)  $x^2 + y^2 \leq 4$  and  $y > x^3$
- (D)  $x^2 + y^2 \leq 4$  and  $y < x^3$

Q.3 A line in the form of  $px + qy = 5$  passes through the points (0, 5) and (2, 1). The gradient ( $m$ ) and  $y$ -intercept ( $b$ ) are:

(A)  $m = 1$                        $b = 2$

(B)  $m = -2$                      $b = 5$

(C)  $m = 5$                        $b = -2$

(D)  $m = 2$                        $b = 1$

Q. 4 The fourth, fifth and sixth terms of an arithmetic series are  $-3$ ,  $5$  and  $13$  respectively. The first term of this series is:

(A) 21

(B)  $-11$

(C)  $-27$

(D) 8

Q. 5 Given  $\log \frac{a}{b} + \log \frac{b}{a} = \log (a + b)$ , then the correct statement below is:

(A)  $a + b = 1$

(B)  $a - b = 1$

(C)  $a = b$

(D)  $a^2 - b^2 = 1$

Q. 6 The function  $f(x)$  is defined by  $f(x) = \begin{cases} \frac{4}{x} & : x > 1 \\ 4^x & : x \leq 1 \end{cases}$ .

What is the value of  $f(0.5) + f(2)$  ?

- (A) 4
- (B) 10
- (C) 18
- (D) 24

Q. 7  $\int \frac{x-4}{x^2} dx$  can be expressed as:

- (A)  $\frac{1}{2} \ln x^2 + 4x + C$
- (B)  $\ln x^2 + \frac{4}{x} + C$
- (C)  $\ln x - 4x + C$
- (D)  $\ln x + \frac{4}{x} + C$

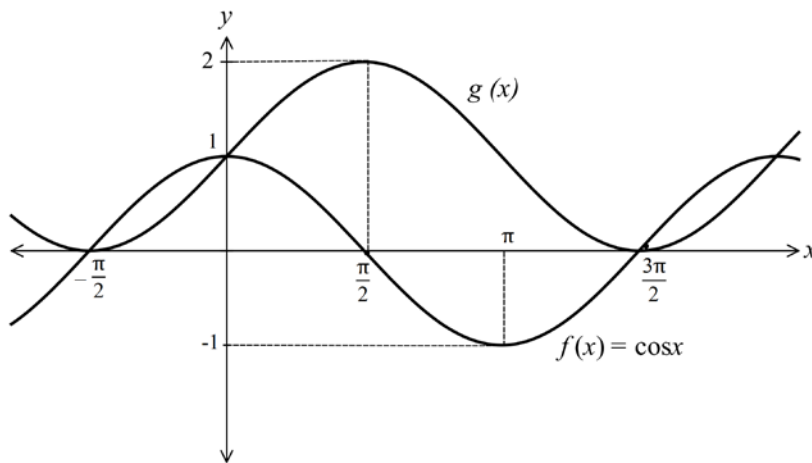
Q. 8 Which of the following statements is true if  $(2, -5)$  is a minimum turning point of  $f(x)$  and  $f(x) = -f(-x)$ ?

- (A)  $(-2, -5)$  is a maximum turning point of  $f(x)$
- (B)  $(-2, 5)$  is a minimum turning point of  $f(x)$
- (C)  $(-2, -5)$  is a minimum turning point of  $f(x)$
- (D)  $(-2, 5)$  is a maximum turning point of  $f(x)$

- Q. 9 The quadratic equation  $x^2 - 9x + 16 = 0$  has roots  $\alpha$  and  $\beta$ .  
The value of  $\sqrt{\alpha} + \sqrt{\beta}$  is:

- (A) 3  
(B)  $\sqrt{13}$   
(C)  $\sqrt{17}$   
(D) 17

- Q. 10 The diagram shows the graphs of  $f(x) = \cos x$  and  $g(x)$ .



The equation of  $g(x)$  is:

- (A)  $g(x) = f\left(x - \frac{\pi}{2}\right) + 2$   
(B)  $g(x) = f\left(x + \frac{\pi}{2}\right) + 1$   
(C)  $g(x) = f\left(x - \frac{\pi}{2}\right) + 1$   
(D)  $g(x) = f\left(x + \frac{\pi}{2}\right) + 2$

**End of Multiple Choice**

**Section II                      90 marks****90 marks****Attempt Questions 11–16****Allow about 2 hours and 45 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations

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**Question 11: Start A New Booklet****15 Marks**

- a. Fully factorise  $16 - 2x^3$ . 2
- b. Solve  $|2x - 5| < 1$ . 2
- c. Evaluate  $\lim_{x \rightarrow -1} \frac{2x^2 + 3x + 1}{x + 1}$  2
- d. Find  $\int_0^{\frac{\pi}{3}} \cos\left(\frac{x}{2}\right) dx$  2
- e. Find the values of  $p$  and  $q$  for which  $p + q\sqrt{6} = \frac{12\sqrt{6}}{\sqrt{6} - 2}$ . 2
- f. Sketch the graph of  $(x - 2)^2 + y^2 = 4$ . 2
- g. Show that  $\frac{d}{dx} \left[ 2x(x - 4)^3 \right] = 8(x - 1)(x - 4)^2$ . 3

**End of Question 11**

**Question 12 Start a New Booklet****15 Marks**

- a. The vertices of  $\triangle ABC$  are A(1, 8) B(7, 4) and C(1,0) as shown below, where AQ is the altitude.

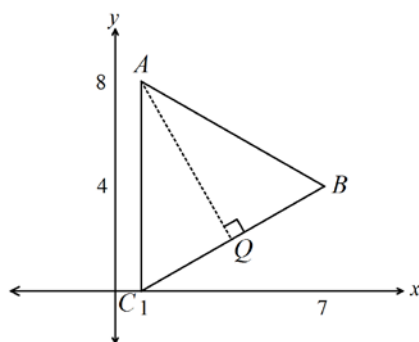


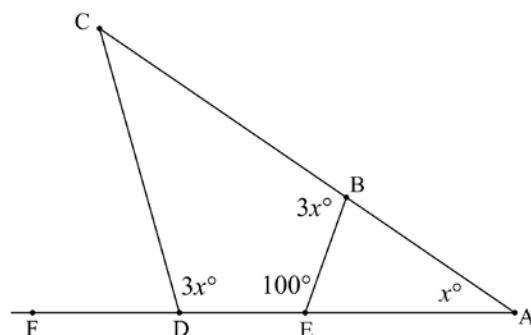
Diagram not to scale

- i. Show the gradient of AQ is  $-\frac{3}{2}$  1
  - ii. Hence, determine the equation of the altitude AQ in general form. 1
  - iii. Hence determine the length of BQ.  
(Leave your answer in exact form) 2
- b. In a geometric series, the fifth term is  $\frac{1}{9}$  and the eight term is  $\frac{1}{243}$ .  
Determine the value of the common ratio (r). 2
- c. What is the value of  $\sum_{m=1}^{\infty} 5\left(\frac{2}{5}\right)^{m-1}$  ? 2
- d. Find  $\int \frac{1}{(2x+3)^5} dx$  . 2

**Question 12 continues on the next page.**

**Question 12 continued:**

e.



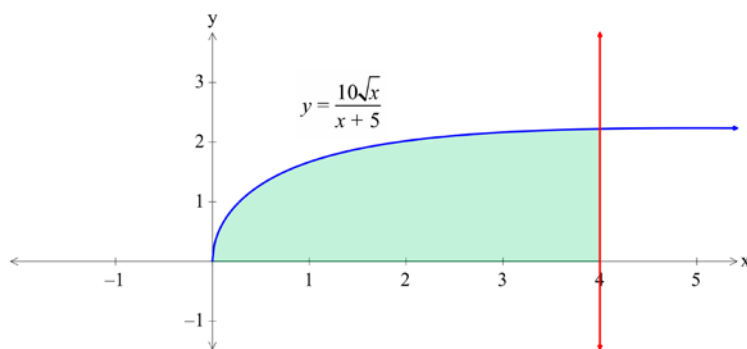
The points B and E lie on the sides AC and AD respectively of  $\triangle ACD$ . The point F lies on AD produced, as shown in the diagram.

(i) Copy the diagram into your answer booklet.  
Make your diagram one third of a page.

(ii) Find the value of  $x$ , giving reasons.

2

f The area bounded by the curve  $f(x) = \frac{10\sqrt{x}}{x+5}$ , the  $x$ -axis and the line  $x = 4$  is shown in the diagram below.



i. Using the Trapezoidal Rule with 5 functional values, determine the approximate area of the shaded region.  
Give your answer correct to two decimal places.

2

ii. Is the estimate calculated in part i) greater or less than the exact area?  
Give a reason to justify your answer.

1

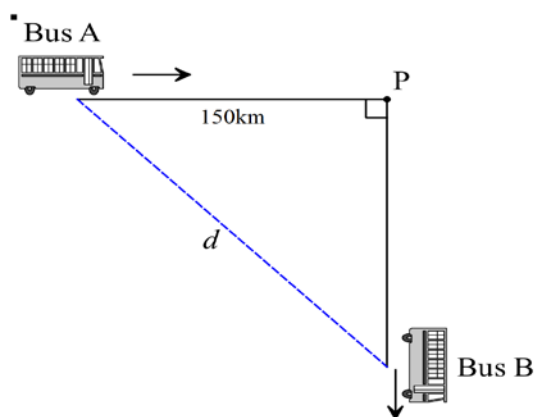
**End of Question 12**

**Question 13 Start A New Booklet****15 Marks**

a. Find  $\int (1 + \tan x) dx$  .

**2**

- b. Two school buses are travelling along straight roads which intersect at right angles at the point P, as shown in the diagram.



Initially, Bus A is 150km due west of P and is travelling towards P at 50km/hr. At the same time Bus B leaves P and travels due south at 40km/hr. Let  $d$  km be the distance between Bus A and Bus B at  $t$  hours after the buses start moving.

- i. Show that Bus A is  $(150 - 50t)$  km from P after  $t$  hours

**1**

- ii. Show that

**2**

$$d = \sqrt{4100t^2 - 15000t + 22500} ,$$

where  $d$  is the distance between the buses when  $0 < t < 3$ .

- ii. Find the value of  $t$  which gives the minimum value of  $d$ .

**2**

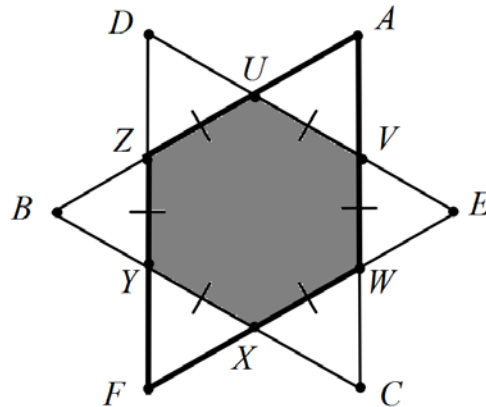
Answer correct to one decimal place.

**Question 13 continues on the next page.**

**Question 13 continued:**

- c. Find the value of the derivative of  $y = 4\tan(2x) - \frac{4x^2}{\pi}$  when  $x = \frac{\pi}{12}$  2

- d. The diagram shows a six pointed star which is drawn using two triangles,  $\triangle ABC$  and  $\triangle FDE$ . The intersection of the two triangles is a regular hexagon.



- i. Show that  $\triangle AVU$  is an equilateral triangle. 2
- ii. Similarly  $\triangle VEW$ ,  $\triangle WCX$ ,  $\triangle XFY$ ,  $\triangle YBZ$  and  $\triangle ZDU$  are all equilateral triangles. 2

Prove that  $ZAWF$  is a rhombus.

- e. Find, in simplest form,  $\frac{d}{dx} \left( \frac{\cos x}{1 - \sin x} \right)$ . 2

**End of Question 13**

**Question 14 Start A New Booklet****15 Marks**

- a. Sketch the graph of  $y = \ln(x - 4)$ , clearly indicating the  $x$ -intercept and any asymptotes. **2**
- b. Consider the curve with equation  $y = 4x^4 + 8x^3 + 1$ .
- Find the stationary points and determine their nature. **4**
  - Show that the point  $(-1, -3)$  is a point of inflexion. **2**
  - Sketch the curve, labelling the stationary points, points of inflexion and  $y$ -intercept. **3**
- c. A particle moves in a straight line such that its displacement  $x$  in metres is given by
- $$x = 70e^{-\frac{t}{10}} - 20t, \text{ where } t \text{ is time in seconds.}$$
- Find the initial displacement of the particle. **1**
  - Will the particle ever come to rest?  
Justify your answer using appropriate calculations. **2**
  - Find the distance travelled by the particle in the first 3 seconds.  
Give the answer correct to 2 decimal places. **1**

**End of Question 14**

**Question 15 Start A New Booklet****15 Marks**

- a. The curve  $f(x)$  has a minimum turning point at  $(2, -10)$ . The second derivative is given by the equation  $f''(x) = 12x - 10$ . Determine the equation of  $f(x)$ . **3**

- b. Determine the value of  $k$  given  $\int_{-2}^2 (x^7 - x + k) dx = 16$  where  $k$  is constant. **2**

- c. The amount of caffeine,  $C(t)$ , in milligrams in your system after drinking a cappuccino is given by

$$C(t) = 105e^{-kt},$$

where  $k$  is a constant and  $t$  is the time in hours that have passed since drinking the cappuccino.

- (i) After one hour the caffeine in your system has decreased by 40%. Find the exact value of  $k$ . **2**
- (ii) When will there be 10 milligrams of caffeine remaining in your system? Give the answer correct to 2 significant figures. **1**
- d. A quadratic function is defined by  $f(x) = x^2 - 2kx + (2k + 3)$ .
- i) Find the values of  $k$  for which the equation  $f(x) = 0$  has two real roots. **2**
- ii) Find the values of  $k$  for which the solutions to  $f(x) = 0$  are both positive. **2**

- d. The rate at which the height of a Jacaranda tree grows is given by

$$\frac{dh}{dt} = \frac{110}{(t + 4)^2} \text{ metres per year,}$$

where  $h$  is the height of the tree in metres and  $t$  is the number of years that have passed since the tree was an established seedling with a height 0.5 m.

- Find the height of the tree when  $t = 5$ , correct to 1 decimal place. **3**

**End of Question 15**

**Question 16 Start A New Booklet****15 Marks**

- a. Penny borrows \$250 000 to be repaid at a reducible interest rate of 0.4% per month. Let  $A_n$  be the amount owing at the end of  $n$  months and  $M$  be the monthly repayment.

(i) Show that  $A_2 = 250\,000(1.004)^2 - M(1+1.004)$  2

(ii) Show that  $A_n = 250\,000(1.004)^n - M\left(\frac{(1.004)^n - 1}{0.004}\right)$  1

(iii) If she repays the amount in 120 months, then show that  $M = \$2627$ , to the nearest dollar. 1

(iv) Penny decides to pay off the loan by making monthly payments of \$3500 instead of \$2627. 3

Show that Penny will make 84 repayments of \$3500 and then a final part payment of \$1001.64.

b. i Show  $\frac{\sec\theta}{\operatorname{cosec}\theta} = \tan\theta$ . 1

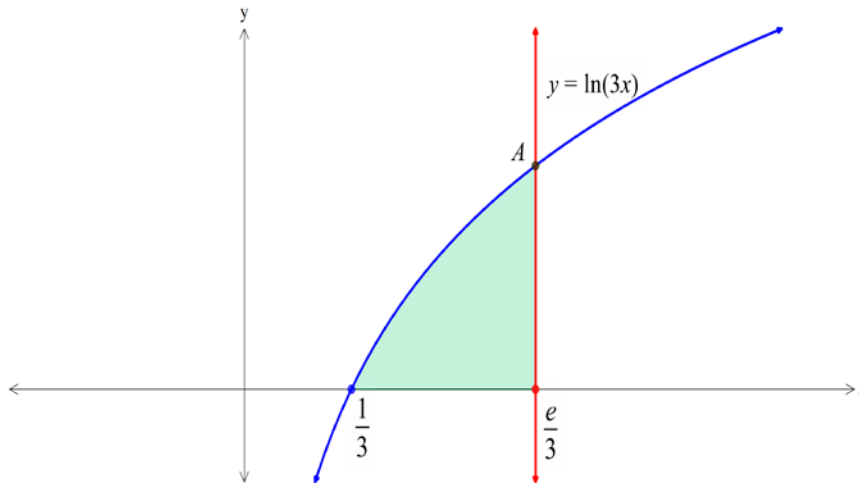
ii Hence, or otherwise, solve  $3\sec^2\left(\frac{x}{2}\right) = \operatorname{cosec}^2\left(\frac{x}{2}\right)$  for  $0 \leq x \leq 2\pi$ . 3

**Question 16 continues on the next page.**

**Question 16 continued:**

- b. The shaded region shown below is rotated around the y-axis.

The point A is the intersection of the curve  $y = \ln(3x)$  and the line  $x = \frac{e}{3}$ .



- |    |                                                               |   |
|----|---------------------------------------------------------------|---|
| i  | Determine the y – value of point A.                           | 1 |
| ii | Determine the exact volume of the solid of revolution formed. | 3 |

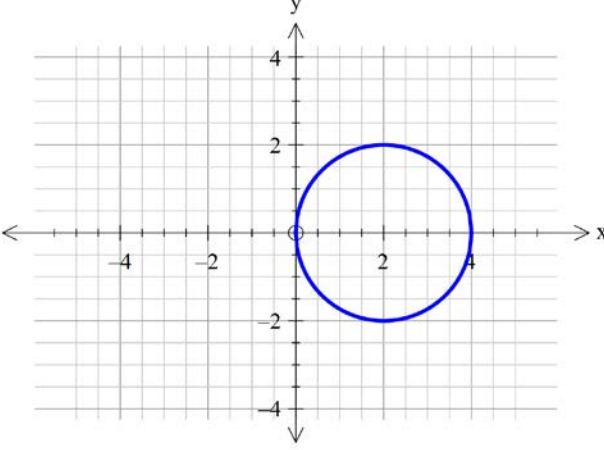
**End of Examination**

## MSC HSC Trial Examination 2017- Solutions

|    |   |                                                                                                                                                                                                                     |
|----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1 | A | $3.61984 \times 10^1 = 3.620 \times 10^1$ (4 sig)                                                                                                                                                                   |
| Q2 | D | Inside the circle and below but not including the cubic.                                                                                                                                                            |
| Q3 | B | sub (0, 5) into $px + qy = 5$<br>$p(0) + q(5) = 5$<br>$5q = 5$<br>$q = 1$<br>$\therefore px + y = 5$<br>sub (2, 1) into $px + y = 5$<br>$p(2) + 1 = 5$<br>$p = 2$<br>$\therefore 2x + y = 5$<br>$m = -2$<br>$b = 5$ |
| Q4 | C | $T_4 = -3$<br>$T_5 = 5$<br>$T_6 = 13$<br>$\therefore d = 8$<br>$T_1 = T_4 - 3d$<br>$= -3 - 3(8)$<br>$= -27$                                                                                                         |
| Q5 | A | $\log\left(\frac{a}{b} * \frac{b}{a}\right) = \log(a + b)$<br>$a + b = 1$                                                                                                                                           |
| Q6 | A | $f(0.5) = 4^{\frac{1}{2}} = 2$ $f(2) = \frac{4}{2} = 2$<br>$f(0.5) + f(2) = 4$                                                                                                                                      |
| Q7 | D | $\int \frac{x}{x^2} - \frac{4}{x^2} dx = \int \frac{1}{x} - 4x^{-2} dx$<br>$= \ln x - \frac{4x^{-1}}{-1} + C$<br>$= \ln x + \frac{4}{x} + C$                                                                        |
| Q8 | D | $f(x) = -f(-x) \Rightarrow f(-x) = -f(x)$<br>$\therefore$ function is odd<br>If minimum turning point at (2, -5) then maximum turning point (-2, 5)                                                                 |
| Q9 | C |                                                                                                                                                                                                                     |

|     |   |                                                                                                                                                                                                                                                                                           |
|-----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |   | $\alpha + \beta = 9$ $\alpha \times \beta = 16$ $(\sqrt{\alpha} + \sqrt{\beta})^2 = (\sqrt{\alpha})^2 + 2\sqrt{\alpha}\sqrt{\beta} + (\sqrt{\beta})^2$ $= \alpha + \beta + 2\sqrt{\alpha\beta}$ $= 9 + 2\sqrt{16}$ $= 9 + 8$ $= 17$ $\therefore \sqrt{\alpha} + \sqrt{\beta} = \sqrt{17}$ |
| Q10 | C | <p>Curve has been lifted by 1 unit vertically and translated <math>\frac{\pi}{2}</math> units to the right therefore</p> $g(x) = f\left(x - \frac{\pi}{2}\right) + 1$                                                                                                                     |

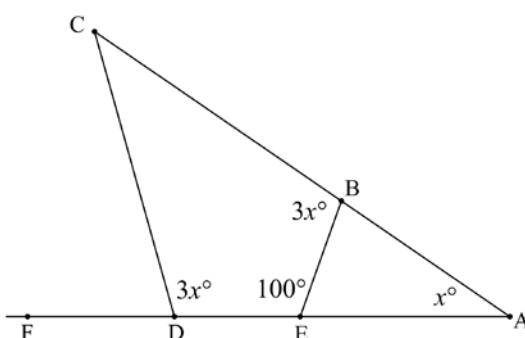
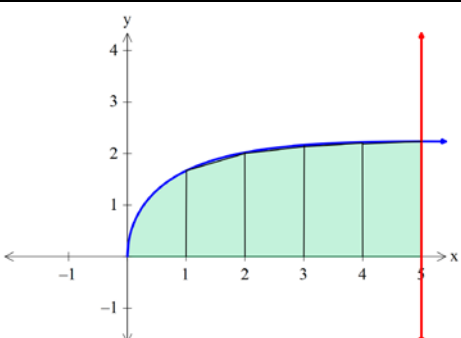
|      |                                                                                                                                                                                                                                          |                                                                                                     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Q11a | $16 - 2x^3 = 2(8 - x^3)$ $= 2(2 - x)(4 + 2x + x^2)$                                                                                                                                                                                      | <p>2 marks for correct solution from correct working</p> <p>1 mark for taking out common factor</p> |
| Q11b | $-1 < 2x - 5 < 1$ $4 < 2x < 6$ $\therefore 2 < x < 3$                                                                                                                                                                                    | 2 marks for correct solution from correct working                                                   |
| Q11c | $\lim_{x \rightarrow -1} \frac{2x^2 + 3x + 1}{x + 1} = \lim_{x \rightarrow -1} \frac{(2x + 1)(x + 1)}{x + 1}$ $= \lim_{x \rightarrow -1} 2x + 1$ $= -2 + 1$ $= -1$                                                                       | 2 marks for correct solution from correct working                                                   |
| Q11d | $\int_0^{\frac{\pi}{3}} \cos\left(\frac{x}{2}\right) dx$ $= 2 \left[ \sin \frac{x}{2} \right]_0^{\frac{\pi}{3}}$ $= 2 \left( \sin \frac{\pi}{6} - \sin 0 \right)$ $= 2 \times \frac{1}{2} = 1$                                           | 2 marks for correct solution from correct working                                                   |
| Q11e | $\frac{12\sqrt{6}}{\sqrt{6} - 2} = \frac{12\sqrt{6}}{\sqrt{6} - 2} \times \frac{\sqrt{6} + 2}{\sqrt{6} + 2}$ $= \frac{72 + 24\sqrt{6}}{6 - 4}$ $= \frac{72 + 24\sqrt{6}}{2}$ $= 36 + 12\sqrt{6}$ $\therefore p = 36 \text{ and } q = 12$ | 2 marks for correct solution from correct working                                                   |

|      |                                                                                                                                                                                                              |                                                   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Q11f |  <p>Centre (2,0) Radius=2</p>                                                                                               | 2 marks for correct graph                         |
| Q11g | $U = 2x \qquad V = (x - 4)^3$ $U' = 2 \qquad V' = 3(x - 4)^2$ $\frac{d}{dx}[2x(x - 4)^3] = VU' + UV'$ $= 2(x - 4)^3 + 6x(x - 4)^2$ $= 2(x - 4)^2[(x - 4) + 3x]$ $= 2(x - 4)^2(4x - 4)$ $= 8(x - 4)^2(x - 1)$ | 3 marks for correct solution from correct working |

Markers Comments

|          |                                                                                                                                                                                                           |                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Q12a(i)  | $m_{AQ} \times m_{CB} = -1$ $m_{AQ} = -\frac{1}{m_{CB}}$ $m_{CB} = \frac{4-0}{7-1} = \frac{2}{3}$ $\therefore m_{AQ} = -\frac{3}{2}$                                                                      | 1 mark – correct demonstration from gradient of CB                                  |
| Q12a(ii) | $y = mx + b$ $y = -\frac{3}{2}x + b$ $x = 1 \Rightarrow y = 8$ $8 = -\frac{3}{2} + b$ $b = 8 + \frac{3}{2} = \frac{19}{2}$ $y = -\frac{3}{2}x + 9.5$ $2y = -3x + 19$ $3x + 2y - 19 = 0$                   | 1 mark – correct formula ie. Both gradient and y-intercept correct                  |
| a-iii    | $D = \frac{ ax + by + c }{\sqrt{a^2 + b^2}}$ $(x,y) \Rightarrow (7,4)$ $3x + 2y - 19 = 0$ $D = \frac{ 3 \times 7 + 2 \times 4 - 19 }{\sqrt{9 + 4}}$ $= \frac{ 10 }{\sqrt{13}}$ $= \frac{10\sqrt{13}}{13}$ | <p>2 marks – correct solution</p> <p>1 mark – correct substitution into formula</p> |

|   |                                                                                                                                                                                                                                                                     |                                                                                                            |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| b | $T_n = ar^{n-1}$ $\frac{T_8}{T_5} = \frac{ar^7}{ar^4} = r^3 = \left(\frac{1}{243}\right) \div \frac{1}{9}$ $r^3 = \frac{1}{27}$ $r = \frac{1}{3}$                                                                                                                   | <p>2 marks – correct solution.</p> <p>1 mark – determining expression for <math>r^3</math></p>             |
| c | $\sum_{m=1}^{\infty} 5 \left(\frac{2}{5}\right)^{m-1}$ - expression is limiting sum therefore<br>$T_1 = a = 5 \left(\frac{2}{5}\right)^0 = 5$ $r = \frac{2}{5}$ $S_{\infty} = \frac{a}{1-r}$ $= \frac{5}{1-\frac{2}{5}}$ $= \frac{5}{\frac{3}{5}}$ $= \frac{25}{3}$ | <p>2 marks – correct solution</p> <p>1 mark – correct first term and common ratio identified.</p> <p>-</p> |
| d | $\int \frac{1}{(2x+3)^5} dx$ $= \int (2x+3)^{-5} dx$ $= \frac{(2x+3)^{-4}}{2 \times -4} + C$ $= -\frac{1}{8(2x+3)^4} + C$                                                                                                                                           | <p>2marks – correct solution</p> <p>1 mark – incorrect denominator</p>                                     |
|   |                                                                                                                                                                                                                                                                     |                                                                                                            |

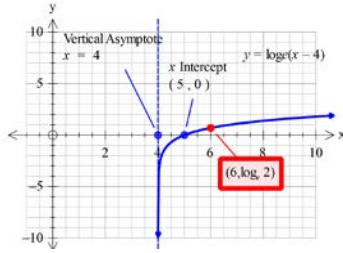
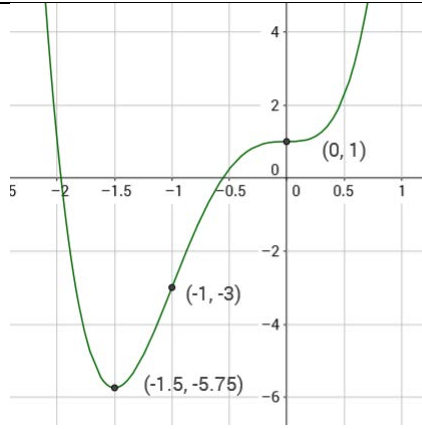
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|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------|----------------|---|---|------------------------------|---|------------------------------|------------------------|----------------------------------------------|----------------|------------------------------------------------------------------------------------|
| e                            | <div></div> <div><math display="block">\angle BEA = 180 - 100 = 80</math><math display="block">\angle CBE = \angle BEA + \angle BAE</math><math display="block">3x = 80 + x</math><math display="block">2x = 80</math><math display="block">x = 40</math></div> <div><div>straight angle</div><div>ext <math>\angle</math> of <math>\Delta = \Sigma</math> opp interior <math>\angle</math></div></div>                                                                                                                                                                                             | <div>2 marks – correct answer with reasons</div> <div>1 mark – correct answer with incomplete reasons</div> |                        |                                              |                |   |   |                              |   |                              |                        |                                              |                |                                                                                    |
| f-i                          | <div><table><tr><td><math>x</math></td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td><math>y = \frac{10\sqrt{x}}{x+5}</math></td><td>0</td><td><math>\frac{10}{6} = \frac{5}{3}</math></td><td><math>\frac{10\sqrt{2}}{7}</math></td><td><math>\frac{10\sqrt{3}}{8} = \frac{5\sqrt{3}}{4}</math></td><td><math>\frac{20}{9}</math></td></tr></table></div> <div><math display="block">A \cong \frac{h}{2} \{ y_0 + 2(y_1 + y_2 + y_3 + y_4) \}</math><math display="block">\cong \frac{1}{2} \times \left( 0 + 2 \left( \frac{5}{3} + \frac{10\sqrt{2}}{7} + \frac{5\sqrt{3}}{4} \right) + \frac{20}{9} \right)</math><math display="block">\cong 6.96</math></div> | $x$                                                                                                         | 0                      | 1                                            | 2              | 3 | 4 | $y = \frac{10\sqrt{x}}{x+5}$ | 0 | $\frac{10}{6} = \frac{5}{3}$ | $\frac{10\sqrt{2}}{7}$ | $\frac{10\sqrt{3}}{8} = \frac{5\sqrt{3}}{4}$ | $\frac{20}{9}$ | <div>2 marks – correct solution</div> <div>1 mark – correct table or similar</div> |
| $x$                          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                                                                                                           | 2                      | 3                                            | 4              |   |   |                              |   |                              |                        |                                              |                |                                                                                    |
| $y = \frac{10\sqrt{x}}{x+5}$ | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\frac{10}{6} = \frac{5}{3}$                                                                                | $\frac{10\sqrt{2}}{7}$ | $\frac{10\sqrt{3}}{8} = \frac{5\sqrt{3}}{4}$ | $\frac{20}{9}$ |   |   |                              |   |                              |                        |                                              |                |                                                                                    |
| f-ii                         | <div></div> <div>As the curve is concave down, the trapeziums would lie under the curve hence estimate is an underestimate.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <div>1 mark – correct answer with reasons.</div>                                                            |                        |                                              |                |   |   |                              |   |                              |                        |                                              |                |                                                                                    |

Markers Comments.

12-a-i a number of students incorrectly interpreted as midpoint

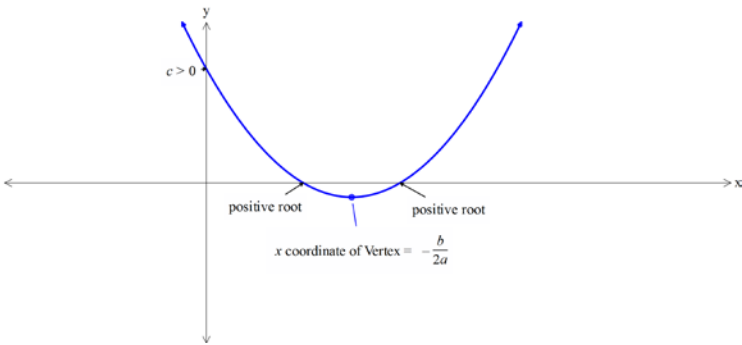
|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                             |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Q13a    | $\int (1 + \tan x) dx = \int \left( 1 + \frac{\sin x}{\cos x} \right) dx$ $= x - \ln \cos x + c$                                                                                                                                                                                                                                                                                                                                                             | 2 marks correct solution from correct working<br>1 mark correct $\tan x = \frac{\sin x}{\cos x}$            |
| Q13b-i  | Distance from P = 150 - distance travelled<br>$= 150 - 50t$                                                                                                                                                                                                                                                                                                                                                                                                  | 1 mark – correct demonstration.                                                                             |
| Q13b-ii | $d = \sqrt{(150 - 50t)^2 + (40x)^2}$ $= \sqrt{22500 - 2 \times 150 \times 50t + 2500t^2 + 1600t^2}$ $= \sqrt{4100t^2 - 15000t + 22500}$                                                                                                                                                                                                                                                                                                                      | 2 marks correct solution from correct working<br>1 mark correct substitution into Pythagoras' Theorem       |
| 13b-iii | As $4100t^2 - 15000t + 22500$ is a parabola with $a = 4100$ , the parabola is concave up and therefore a minimum.<br><br>$f'(t) = 8200t - 15000$ stationary point at $f'(t) = 0$<br>$0 = 8200t - 15000$ $t = \frac{15000}{8200}$ $= 1.82926.....$ $= 1.8$                                                                                                                                                                                                    | 2 marks correct solution from correct working.<br>1 marks correct x with no justification for minimum value |
| 13c     | $y = 4\tan(2x) - \frac{4x^2}{\pi}$ $y' = 4 \times \sec^2(2x) \times 2 - \frac{8x}{\pi}$ $= 8 \left( \sec^2(2x) - \frac{x}{\pi} \right)$ at $x = \frac{\pi}{12}$<br>$y' = 8 \left( \sec^2\left(\frac{2\pi}{12}\right) - \frac{\frac{\pi}{12}}{\pi} \right)$ $= 8 \left( \sec^2\left(\frac{\pi}{6}\right) - \frac{1}{12} \right)$ $= 8 \left( \left(\frac{2}{\sqrt{3}}\right)^2 - \frac{1}{12} \right)$ $= 8 \left( \frac{4}{3} - \frac{1}{12} \right)$ $= 10$ | 2 marks correct solution from correct working.<br>1 mark correct derivative.                                |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                       |
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| 13di  | <p>ZUVWXY regular hexagon (given)</p> $\text{Interior angle} = \frac{180(6-2)}{6}$ $= 120^\circ$ $\angle AUV = \angle AVU$ $= 180^\circ - 120^\circ \text{ (straight } \angle \text{)}$ $= 60^\circ$ $\angle VAU = 180^\circ - (60^\circ + 60^\circ) \text{ (}\angle \text{ sum of } \Delta \text{)}$ $= 60^\circ$ $\therefore \Delta AVU \text{ is equilateral}$                                                                              | <p>2 marks correct solution from correct working with reasons.</p> <p>1 mark finding an interior angle of hexagon</p> |
| 13dii | $\angle ZAW = \angle ZFW$ $= 60^\circ \text{ (equilateral } \Delta s \text{)}$ $\angle AZF = \angle AWF$ $= 120^\circ \text{ (}\angle \text{ of regular hexagon)}$ $\therefore ZAWF \text{ is a parallelogram (opposite } \angle = \text{)}$ $AU = AV \text{ (}\Delta AVU \text{ equilateral)}$ $ZU = VW \text{ (given)}$ $\therefore ZA = WA$ $\therefore ZAWF \text{ is a rhombus}$ $\text{(parallelogram with a pair of adjacent sides =)}$ | <p>2 marks correct solution from correct working with reasons.</p> <p>1 mark proving ZAWF is a parallelogram</p>      |
| 13e   | $\frac{d}{dx} \left( \frac{\cos x}{1 - \sin x} \right)$ $u = \cos x \quad v = 1 - \sin x$ $u' = -\sin x \quad v' = -\cos x$ $\frac{d}{dx} \left( \frac{\cos x}{1 - \sin x} \right) = \frac{-\sin x(1 - \sin x) - \cos x(-\cos x)}{(1 - \sin x)^2}$ $= \frac{-\sin x + \sin^2 x + \cos^2 x}{(1 - \sin x)^2}$ $= \frac{1 - \sin x}{(1 - \sin x)^2}$ $= \frac{1}{1 - \sin x}$                                                                     | <p>2 marks correct solution from correct working</p> <p>1 mark correct u' and v'</p>                                  |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                             |    |      |    |   |   |    |     |   |   |   |    |       |   |   |   |   |   |                                                                                                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|----|---|---|----|-----|---|---|---|----|-------|---|---|---|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q14a) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>2 marks correct answer with 2<sup>nd</sup> pt</p> <p>1 mark correct asymptote and x-int</p>                                                                              |    |      |    |   |   |    |     |   |   |   |    |       |   |   |   |   |   |                                                                                                                                                                           |
| bi)   | $y' = 16x^3 + 24x^2$ $y' = 0 \text{ for a S.P.}$ $8x^2(2x + 3) = 0$ $x = 0 \text{ and } x = -\frac{3}{2}$ $y = 1 \text{ and } y = -\frac{23}{4} \text{ respectively}$ <p>Using a slope table</p> <table border="1" data-bbox="292 792 940 983"><tr><td>x</td><td>-2</td><td>-1.5</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>y'</td><td>-32</td><td>0</td><td>8</td><td>0</td><td>40</td></tr><tr><td>slope</td><td>\</td><td>-</td><td>/</td><td>-</td><td>/</td></tr></table> <p><math>(-\frac{3}{2}, -\frac{23}{4})</math> is a minimum and <math>(0,1)</math> a horizontal point of inflexion</p> <p>Or use of <math>y''</math> to show concavity change<br/><math>y'' = 0</math> is not sufficient for a point of inflection</p> | x                                                                                                                                                                           | -2 | -1.5 | -1 | 0 | 1 | y' | -32 | 0 | 8 | 0 | 40 | slope | \ | - | / | - | / | <p>4 marks: correct answer</p> <p>3 marks: correct coordinates and their nature, showing HPOI</p> <p>2 marks: correct co-ords</p> <p>1 mark: correct first derivative</p> |
| x     | -2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | -1.5                                                                                                                                                                        | -1 | 0    | 1  |   |   |    |     |   |   |   |    |       |   |   |   |   |   |                                                                                                                                                                           |
| y'    | -32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0                                                                                                                                                                           | 8  | 0    | 40 |   |   |    |     |   |   |   |    |       |   |   |   |   |   |                                                                                                                                                                           |
| slope | \                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                                                                                                                                                                           | /  | -    | /  |   |   |    |     |   |   |   |    |       |   |   |   |   |   |                                                                                                                                                                           |
| ii)   | $y'' = 48x^2 + 48x$ $y'' = 0 \text{ for possible points of inflexion}$ $48x(x + 1) = 0$ $x = -1, 0$ <p>or sub <math>x=1</math> into 2<sup>nd</sup> derivative<br/>this point of inflection is NOT a horizontal point of inflection</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>2 marks: correct answer</p> <p>1 mark: correct second derivative</p>                                                                                                     |    |      |    |   |   |    |     |   |   |   |    |       |   |   |   |   |   |                                                                                                                                                                           |
| iii)  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>3 marks: correct answer</p> <p>2 marks: correct shape</p> <p>1 mark correct stationary points</p> <p>Cfe applicable</p> <p>NOTE some very poor graphs were presented</p> |    |      |    |   |   |    |     |   |   |   |    |       |   |   |   |   |   |                                                                                                                                                                           |

|      |                                                                                                                                              |                       |
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| c)i  | $x = 70e^0 - 20(0)$<br>$x = 70\text{m}$                                                                                                      | 1 mark correct answer |
| ii)  | $= -7e^{\frac{t}{10}} - 20$<br>$= 0$ for particle at rest<br>$e^{-\frac{t}{10}} \neq -\frac{20}{7}$<br>since $e^{-\frac{t}{10}} > 0$ for all |                       |
| iii) | from i) $t=0$ and $x=70$<br>$t=3$ $x=-8.14$<br>therefore distance travelled is 78.14m                                                        | 1 mark correct answer |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                      |
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| Q15<br>a    | $f''(x) = 12x - 10$<br>$f'(x) = 6x^2 - 10x + c_1$<br>stationary points at $f'(x) = 0$ when $x = 2$<br>$f'(2) = 6(2)^2 - 10(2) + c_1 = 0 \Rightarrow c_1 = -4$<br>$\therefore f'(x) = 6x^2 - 10x - 4$<br>$f(x) = 2x^3 - 5x^2 - 4x + c_2$<br>sub point $(2, -10)$<br>$-10 = 2(2)^3 - 5(2)^2 - 4(2) + c_2 \Rightarrow c_2 = 2$<br>$\therefore f(x) = 2x^3 - 5x^2 - 4x + 2$                                                                                                                                        | 3 marks correct solution from correct working<br><br>2 marks- partial correct/ one error<br><br>1 mark- partial correct with one required process                    |
| Q15<br>b    | Method 1<br>$\int_{-2}^2 x^7 - x + k \, dx = \left[ \frac{x^8}{8} - \frac{x^2}{2} + kx \right]_{-2}^2$ $= \left( \frac{256}{8} - \frac{4}{2} + 2k \right) - \left( \frac{256}{8} - \frac{4}{2} - 2k \right)$ $= 4k$ $\therefore \quad \begin{aligned} 4k &= 16 \\ k &= 4 \end{aligned}$<br>Method 2<br>$\int_{-2}^2 (x^7 - x + k) \, dx$ $= \int_{-2}^2 (x^7 - x) \, dx + \int_{-2}^2 k \, dx$ $= 0 + [kx]_{-2}^2$<br>(as first function is an odd function)<br>$\therefore 2k - (-2k) = 16$ $4k = 16$ $k = 4$ | 2 marks correct solution from correct working<br><br>1 mark for integration                                                                                          |
| Q15<br>c(i) | when $t = 0$ $C(0) = 105$<br>when $t = 1$ $C(1) = 0.6 \times 105 = 63$ and $C(1) = 105e^{-k}$<br>$\therefore 105e^{-k} = 63$<br>$e^{-k} = \frac{63}{105}$<br>$-k = \ln\left(\frac{63}{105}\right)$<br>$k = -\ln(0.6)$                                                                                                                                                                                                                                                                                          | 2 marks correct solution from correct working<br><br>1 mark for using 0.4 in equation and solving with correct process<br><br>Common error: Using 40% instead of 60% |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <p>Q15<br/>c(ii)</p> | $105e^{-kt} = 10$ $e^{-kt} = \frac{10}{105} = \frac{2}{21}$ $-kt = \ln\left(\frac{2}{21}\right)$ $t = \ln\left(\frac{2}{21}\right) \div \ln(0.6)$ $t \approx 4.6 \text{ hours (2 sigfig)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>1 marks correct solution from correct working</p> <p>1 mark if error carried from part i) showing correct process</p> <p>Common error: Rounding to 3 sig fig (4.60)</p>                                                                                                                                                                                                                                                |
| <p>Q15<br/>d(i)</p>  | <p>For 2 real roots <math>\Delta &gt; 0</math></p> $\Delta = b^2 - 4ac$ $= (-2k)^2 - 4(2k + 3)$ $= 4k^2 - 8k - 12$ $4k^2 - 8k - 12 > 0$ $k^2 - 2k - 3 > 0$ $(k + 1)(k - 3) > 0$ $\therefore k < -1 \text{ or } k > 3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>2 marks correct solution from correct working</p> <p>1 mark for discriminant</p>                                                                                                                                                                                                                                                                                                                                       |
| <p>Q15<br/>d(ii)</p> | <p>Method 1</p> <p>For there to be two roots, <math>k</math> must satisfy <math>k &lt; -1</math> or <math>k &gt; 3</math></p> <p>Since <math>a &gt; 0</math>, parabola is concave up, for there to be two positive roots <math>k</math> must satisfy additional conditions of constant term <math>&gt; 0</math> and <math>-\frac{b}{2a} &gt; 0</math> - see diagram below.</p>  <p><math>\therefore c &gt; 0</math><br/>i.e. <math>2k + 3 &gt; 0</math></p> $k > -\frac{3}{2}$ <p>Also, axis of symmetry <math>x = -\frac{b}{2a}</math> must be <math>&gt; 0</math></p> $x = -\frac{b}{2a} = \frac{2k}{2(1)} = k$ <p><math>\therefore k &gt; 0</math></p> <p>Therefore to satisfy all conditions, <math>k</math> must simultaneously satisfy</p> | <p>2 marks correct solution from correct working</p> <p>1 mark for considering one condition</p> <p>Common error:<br/>Not finding the other condition of <math>k &gt; 0</math></p> <p>Using the quadratic equation and then solving for <math>x &gt; 0</math> resulted in algebraic error<br/>(If no error, successfully found <math>k &gt; -3/2</math> condition but failed to find <math>k &gt; 0</math> condition)</p> |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                  |
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|          | $k > -\frac{3}{2}, k > 0 \text{ and } k > 3$<br>therefore $k > 3$ satisfies all conditions.<br><br>Method 2<br>$\alpha + \beta > 0$ and $\alpha \beta > 0$<br>$\alpha + \beta = -\frac{b}{a} = 2k$<br>$\therefore 2k > 0 \Rightarrow k > 0$<br><br>$\alpha \beta = \frac{c}{a} = 2k + 3$<br>$\therefore 2k + 3 > 0 \Rightarrow k > -\frac{3}{2}$<br><br>Considering all conditions $k > 3$                                                                      |                                                                                                                                                                                                                                                                                  |
| Q15<br>e | Method 1<br>$h = \int_0^5 110(t+4)^{-2} dt + 0.5$ $= 110 \left[ \frac{(t+4)^{-1}}{-1} \right]_0^5 + 0.5$ $= 110 \left[ -\frac{1}{t+4} \right]_0^5 + 0.5$ $= 110 \left[ -\frac{1}{9} + \frac{1}{4} \right] + 0.5$ $\approx 15.8m$<br>Method 2<br>$h = \int 110(t+4)^{-2} dt$ $= -\frac{110}{t+4} + c$ when $t = 0, h = 0.5$<br>$0.5 = -\frac{110}{4} + c$ $c = 28$ $\therefore h = -\frac{110}{t+4} + 28$ when $t = 5$<br>$h = -\frac{110}{9} + 28 \approx 15.8$ | 3 marks correct solution from correct working<br><br>2 marks for definite integral but omitting the 0.5<br><br>1 mark for integration<br><br><br><br><br>3 marks correct solution from correct working<br><br>2 marks for integration and constant<br><br>1 mark for integration |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                     |
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| Question 16 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                     |
| 16 a i)     | $P = \$ 250\,000, R = 1.004$ $A_n = A_{n-1}R - M$ $\Rightarrow A_1 = 250\,000(1.004) - M$ $A_2 = A_1R - M$ $\Rightarrow A_2 = [250\,000(1.004) - M](1.004) - M$ $A_2 = 250\,000(1.004)^2 - 1.004M - M$ $A_2 = 250\,000(1.004)^2 - M(1 + 1.004)$                                                                                                                                                                                                                                                                             | <p>2 Marks: Correct solution</p> <p>1 Mark: correct expression for <math>A_1</math></p>                             |
| 16 a ii)    | $A_n = 250\,000(1.004)^n - M(1 + 1.004 + \dots + 1.004^{n-1})$ $S_n = a \left[ \frac{R^n - 1}{R - 1} \right] \text{ and } a = 1, R = 1.004$ $= 250\,000(1.004)^n - M \left[ \frac{1.004^n - 1}{0.004} \right]$                                                                                                                                                                                                                                                                                                              | <p>1 Mark: Correct solution</p> <p>Must show series up to and including <math>1.004^{n-1}</math></p>                |
| 16 a iii)   | $A_{120} = 250\,000(1.004)^{120} - M \left[ \frac{1.004^{120} - 1}{0.004} \right]$ $0 = 250\,000(1.004)^{120} - M(153.631\dots)$ $M = \frac{250\,000(1.004)^{120}}{153.631\dots}$ $M = 2627.2655\dots \approx \$2627$                                                                                                                                                                                                                                                                                                       | <p>1 Mark: Correct solution</p>                                                                                     |
| 16 a iv)    | $A_n = 250\,000(1.004)^n - 3500 \left[ \frac{1.004^n - 1}{0.004} \right]$ $0 = 250\,000(1.004)^n - 875\,000(1.004^n - 1)$ $0 = 250\,000(1.004)^n - 875\,000(1.004)^n + 875\,000$ $0 = -625\,000(1.004)^n + 875\,000$ $(1.004)^n = \frac{875\,000}{625\,000}$ $n \log_e(1.004) = \log_e \left( \frac{875}{625} \right)$ $n = \log_e \left( \frac{875}{625} \right) \div \log_e(1.004) = 84.286\dots$ <p><math>\therefore</math> 84 full payments and part payment of<br/> <math>0.286\dots(\\$3500) = \\$ 1001.64</math></p> | <p>3 Marks: Correct solution.</p> <p>2 Marks: Correct derivation of <math>n = 84.286\dots</math></p> <p>1 Mark:</p> |
| 16 b i)     | $\frac{\sec\theta}{\csc\theta} = \left( \frac{1}{\cos\theta} \right) \div \left( \frac{1}{\sin\theta} \right) = \frac{\sin\theta}{\cos\theta} = \tan\theta$                                                                                                                                                                                                                                                                                                                                                                 | <p>1 Mark: Correct solution</p>                                                                                     |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
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| 16 b ii) | $3\sec^2\left(\frac{x}{2}\right) = \operatorname{cosec}^2\left(\frac{x}{2}\right) \Rightarrow \frac{3}{\cos^2\left(\frac{x}{2}\right)} = \frac{1}{\sin^2\left(\frac{x}{2}\right)}$ $\frac{\sin^2\left(\frac{x}{2}\right)}{\cos^2\left(\frac{x}{2}\right)} = \frac{1}{3} \Rightarrow \tan^2\left(\frac{x}{2}\right) = \frac{1}{3}$ $\tan\left(\frac{x}{2}\right) = \pm \frac{1}{\sqrt{3}}$ $0 \leq x \leq 2\pi$ $\Rightarrow 0 \leq \frac{x}{2} \leq \pi \Rightarrow \frac{x}{2} = \frac{\pi}{6}, \frac{5\pi}{6} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$ |  |
| 16 c i)  | $x = \frac{e}{3} \rightarrow y = \log_e(3x) \Rightarrow y = \log_e\left(3 \cdot \frac{e}{3}\right) = \log_e(e) = 1$                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
| 16 c ii) | $V_1 = \text{cylinder} = \pi r^2 h = \pi \left(\frac{e}{3}\right)^2 \cdot 1 = \frac{\pi e^2}{9}$ $V_2 = \text{solid of revolution} = \pi \int_0^1 x^2 dy$ $y = \log_e(3x) \rightarrow 3x = e^y \rightarrow x^2 = \left(\frac{e^y}{3}\right)^2 = \frac{e^{2y}}{9}$ $V_2 = \left(\frac{\pi}{9}\right) \int_0^1 e^{2y} dy = \frac{\pi}{9} \left[\frac{1}{2} e^{2y}\right]_0^1 = \frac{\pi}{18} [e^2 - 1]$ $V = V_1 - V_2 = \frac{\pi e^2}{9} - \frac{\pi}{18} [e^2 - 1]$ $V = \frac{\pi(e^2 + 1)}{18}$                                                            |  |